MIL-S-1484E
31 March 1986
SUPERSEDING
MIL-S-1484D
4 October 1979

MILITARY SPECIFICATION

SHIELDS, STOVEPIPE, TENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers two types of stovepipe tent shields.

1.2 Classification. Stovepipe tent shields shall be of the following types as specified (see 6.2):

Type I - Round hole
Type II - Elliptical hole

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-P-268	- Paper, Kraft, Wrapping
PPP-B-636	- Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8340

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MILITARY

MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing
FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-147 - Palletized Unit Loads

DRAWING

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-4-120 - Shields, Stovepipe, Tent

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following document of the issue in effect on date of invitation for bids or request for proposal forms a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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* 2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

* 3.2 Materials. The materials shall conform to the applicable documents referenced herein. Materials not definitely specified shall be of the quality normally used in the manufacture of items of this type. Recycled material may be used (see 6.4).

3.2.1 Steel, sheet. The sheet steel shall be nominal 0.0299-inch thick vitreous enameling iron sheet.

3.2.2 Enamel, vitreous. Vitreous enamel shall be of a type fired at approximately 1500°F. The color of the enamel shall be within the range of color No. 34079 (Forest Green) to color No. 34087 (Olive Drab) conforming to FED-STD-595.

3.3 Construction. The construction of the shields shall conform to the requirements specified herein and as shown on Drawing 5-4-120.

3.3.1 Types I and II shields. Type I and II shields shall be fabricated of steel specified in 3.2.1.

3.3.2 Welding. All welding shall be complete, sound, and free from burn through areas, pits, cracks, or fractures.

3.3.3 Removal of rough or sharp edges, burrs or slivers. All rough or sharp edges, burrs, or slivers shall be removed before enameling.

3.4 Finish.

3.4.1 Enamel, vitreous. After complete fabrication, all surfaces of the shield shall be coated with vitreous enamel specified in 3.2.2. The coating shall level out to an adherent, continuous uniform film and shall be free from scratch, dig, or abrasion exposing bare metal, or area of corrosion. Evidence of cracking in the build-up of vitreous enamel in the recessed areas is allowed. The gloss of the heat treated vitreous enamel shall be no more than 15 when tested as specified in 4.4.5.

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3.4.1.1 Thickness. When tested as specified in 4.4.5, the average vitreous enamel coating thickness shall be within the limits of a minimum of 0.004 inch and a maximum of 0.008 inch in thickness.

3.4.1.2 Adherence. After the shield has been struck with a hammer during the adherence of the vitreous enamel coating test as specified in 4.4.5, the area of exposed metal shall be not more than one-half of the deformed area on the concave side of the indentation.

3.4.1.3 Resistance to heat. There shall be no flaking, cracking, or deterioration of the vitreous enamel coating except for permissible slight change in color when tested as specified in 4.4.5.

3.4.1.4 Resistance to thermal shock. There shall be no flaking, cracking, or deterioration of the vitreous enamel coating when tested as specified in 4.4.5.

3.5 Marking for identification. Each shield shall be permanently and distinctly marked for identification in accordance with MIL-STD-130. The markings shall include the letters "US" and the manufacturer's identification. The markings shall be embossed prior to enameling and shall be located as shown on Drawing 5-4-120.

3.6 Replacement of defective components. During the manufacturing process, components having material defects or damages that are classified as defects in 4.4.3 and 4.4.4 shall be removed from production and replaced with nondefective and properly matched components.

3.7 Workmanship. The finished shields shall be clean and free from damage, cracks, breaks, malformation, oil, grease, or dirt and shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable quality level.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3. First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4, and tested as specified in 4.4.5. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be made in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of manufacturing to determine whether operations or assemblies are accomplished as specified. Inspection shall be conducted to determine that the welding is being performed in accordance with 3.3.2 and Drawing 5-4-120. Whenever nonconformance is noted, correction shall be made to the affected items and the operation. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance.

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* 4.4.3 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of shields. The sample unit shall be one shield. The inspection level shall be S-1, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects and 10.0 for major and minor defects combined.

TABLE I. Classification of visual defects

Examine	Defect	Classification	
		Major	Minor
Finish	Not finished	101	
	Color of finish is not as specified		201
	Coating not adherent (e.g., blistered, peeled, or flaky)	102	
	Coating is not continuous or uniform (i.e., area of no film or not coated)		202
	Scratch, dig, or abrasion exposing bare metal		203
	Area of corrosion	103	
	NOTE: Evidence of cracking in the build-up of vitreous enamel in recessed areas will not be classified as a defect.		
Construction and workmanship	Shield not fabricated of vitreous enameling sheet material	104	
	Not fabricated as specified	105	
	Not clean (i.e., evidence of oil, grease, or dirt)		204
	Damage, crack, break, or malformed	106	
	Rough or sharp edge, burr, or sliver		205
Marking for identification	Missing, illegible, misspelled, or incorrect		206
	Not located as shown on drawing		207
	Not embossed		208

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* 4.4.4 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified on Drawing 5-4-120. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of shields. The sample unit shall be one shield. The inspection level shall be S-2, and the AQL, expressed as defects per hundred units, shall be 4.0.

4.4.5 End item testing. Testing of the completely fabricated shields shall be performed in accordance with tests indicated in table II for the characteristics shown therein. The lot size shall be expressed in units of shields. The sample unit for testing as specified in 4.5.1, 4.5.2, and 4.5.3 shall be one shield. The sample unit for testing as specified in 4.5.4 shall be one shield, as a separate shield is required for this test. All test reports shall contain the individual values used to express the final result. The sample size shall be as follows:

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II. End item tests

Characteristic	Requirement paragraph	Test method	No. of determinations per sample unit	Results reported as
Gloss	3.4.1	4.5.5	1	Pass or fail
Thickness	3.4.1.1	4.5.1	3	Average of 3 determinations to nearest 0.001 inch
Adherence	3.4.1.2	4.5.2	1	Pass or fail
Heat resistance	3.4.1.3	4.5.3	1	Pass or fail
Thermal shock resistance	3.4.1.4	4.5.4	1	Pass or fail

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4.4.6 Packaging inspection. An inspection shall be made to determine that packing and marking comply with section 5 requirements. Defects shall be scored in accordance with table III. The lot size shall be the number of shipping containers in the inspection lot. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

TABLE III. Packaging inspection

Examine	Defect
Marking (exterior)	Omitted; incorrect; illegible; of improper size location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete closure or container flap, loose strapping, inadequate stapling, or improper taping Bulged or distorted container
Content	Number of shields per shipping container is more or less than required 1/

1/ For this defect, one shipping container in the sample shall be examined.

* 4.4.7 Palletization inspection. An examination shall be made to determine that palletization complies with the section 5 requirements. Defects shall be scored as specified in table IV. The lot size shall be the number of palletized unit loads submitted for inspection at one time. The sample unit shall be one palletized unit load ready for shipment. The inspection level shall be S-1, and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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TABLE IV. Palletization inspection

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Enamel coating thickness. The enamel coating thickness shall be measured by a dry magnetic type gage or any other industrial method acceptable to the Government to determine compliance with the enamel coating thickness requirement in 3.4.1.1.

4.5.2 Adherence of enamel coating. The shield shall be struck with the round head of a ball-peen hammer sufficiently hard to cause permanent indentation. The exposed metal area and the deformed area on the concave side shall be measured to determine compliance with the adherence of enamel coating requirement in 3.4.1.2.

4.5.3 Heat resistance of enamel coating. The shield shall be placed in an atmosphere of 1000°F and allowed to remain therein for 20 minutes. At the end of the 20-minute exposure period at 1000°F, the shield shall be removed and allowed to cool to room temperature. After cooling, the enamel coating of the shield (except hammered areas previously tested - see 4.5.2) shall be examined visually to determine compliance with the resistance to heat requirement specified in 3.4.1.3.

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4.5.4 Thermal shock resistance of enamel coating. The shield shall be placed in an atmosphere of 750°F and allowed to remain therein for 10 minutes. At the end of the 10-minute exposure period at 750°F, the shield shall be immediately immersed in water at a temperature of 50°F to 70°F. After cooling in the water, the enamel coating of the shield shall be examined visually to determine compliance with the resistance to thermal shock requirement specified in 3.4.1.4.

4.5.5 Gloss. The gloss shall be measured in accordance with Method 6101 of FED-STD-141 to determine compliance with the gloss requirement in 3.4.1.

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.1.1 Level A packing. A quantity not to exceed sixty tent shields of one type only shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, V2s of PPP-B-636. The inside of each shipping container shall be fitted with a liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. The shields shall be packed on edge in one row within the shipping container and separated one from the other with a slipsheet having a 30 pound minimum basis weight conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.1.2 Level B packing. A quantity not to exceed sixty tent shields of one type only shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF, variety optional, or type SF, class domestic of PPP-B-636. The inside of each shipping container shall be fitted with a liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. The shields shall be packed on edge in one row within the shipping container and separated as specified in 5.1.1. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.1.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

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* 5.1.3 Commercial packing. Shields shall be packed in accordance with ASTM D 3951.

* 5.2 Palletization. When specified (see 6.2), stovepipe shields, packed as specified in 5.1, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size that does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

* 5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The shields are for protective insulation against excessive stove heat where the stovepipe passes through the roof, side, or end of a canvas tent. The type I shield having the round hole is intended to be used where the opening for the stovepipe is through a vertical side or end of tent. The type II shield having an elliptical hole is intended to be used where the opening for the stovepipe is through a slanted roof of tent.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type required (see 1.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Selection of applicable level of packing (see 5.1).
e. Type and class of unit load required (see 5.1.1).
f. Whether weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.1.2.1).
g. When palletization is required (see 5.2).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The first article should consist of two finished shields. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for inspection and approval of the first article.

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6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

6.5 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-1484E	Shields, Stovepipe, Tent

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OI, 1 Apr 84

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